

REVIEW · METABOLIC MEDICINE

When Ozempic stops, the body's old settings start to return

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27 Aug 2026

Semaglutide leaves slowly, but its appetite, weight, and glucose effects usually do not last on their own. Most people regain at least some weight; blood sugar and other risk markers often drift back; nausea and other stomach effects usually ease.

In a follow-up to the STEP 1 trial, 327 adults stepped off weekly semaglutide after 68 weeks. They had lost an average of 17.3% of their body weight. One year later, they had regained 11.6 percentage points - roughly two-thirds of that loss - yet they were still 5.6% lighter than when they began.¹ The graph did not show a trapdoor. It showed a slope.

That slope is the story of stopping Ozempic. Semaglutide is not known to produce a classic chemical withdrawal syndrome. Instead, the drug's effects fade and the biology it had been holding back becomes audible again: hunger, food reward, glucose production, and the body's persistent defense of its prior weight. One caveat matters. Ozempic is semaglutide prescribed for type 2 diabetes at lower doses than the 2.4 mg obesity regimen sold as Wegovy; most direct withdrawal evidence comes from the higher-dose obesity trials, while diabetes-specific stopping data are thinner.^{2,3}

01 The last injection keeps working for weeks

Semaglutide has a biological half-life of about seven days. After the final dose, its concentration falls by roughly half each week, so meaningful exposure fades over about five weeks rather than ending the next morning.⁴ There is no single day on which the body flips from "on Ozempic" to "off Ozempic."

While present, this GLP-1 receptor agonist makes eating easier to regulate. In controlled studies, semaglutide reduced hunger and food cravings, increased fullness, and cut freely chosen meal energy by about 24% to 39%.^{5,6,7,8} A 60-week trial found that people taking 2.4 mg still ate about 240 to 292 fewer calories at laboratory meals than those taking placebo, even after some subjective appetite differences had faded.⁹ As drug exposure falls, that pharmacological brake weakens. Hunger or persistent thoughts about food can return, and maintaining the same intake may require more conscious effort.

The sequence and its evidence boundaries are summarized in [Figure 1](#).

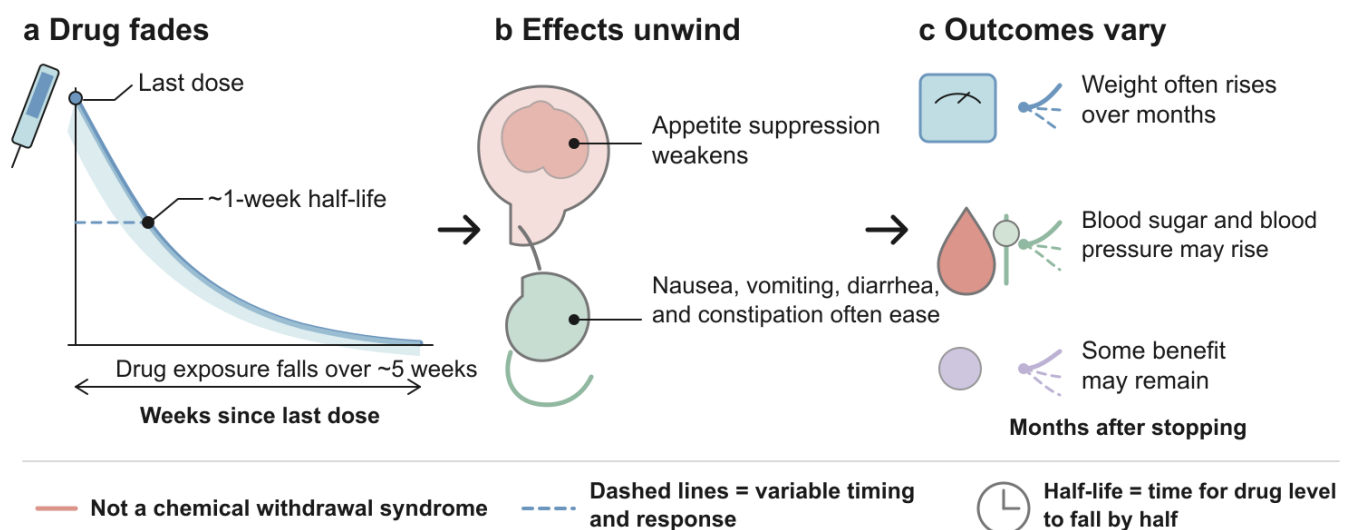


Figure 1. The drug fades first; appetite and metabolic protection follow. Read from left to right: semaglutide exposure falls over several weeks, the appetite brake weakens, stomach side effects often ease, and weight and metabolic markers may then move toward baseline. The exact timing varies, and the final panel describes population averages rather than an individual forecast.^{4,6,10,11}

02 The scale usually turns, but not all the way at once

The most direct experiment was STEP 4. After a 20-week semaglutide run-in, 803 adults were randomized either to continue 2.4 mg or switch to placebo while both groups kept receiving lifestyle support. Over the next 48 weeks, continuers lost another 7.9%; switchers gained 6.9%.¹⁰ Lifestyle counseling did not fully replace the drug's appetite effect.

The broader literature points the same way, but the exact pace depends on how much weight was lost, which drug and dose were used, and how long researchers watched.

EVIDENCE	PEOPLE	OFF-DRUG INTERVAL	MAIN FINDING
STEP 4 ¹⁰	803	48 weeks	Switchers gained 6.9%
STEP 1 extension ¹	327	1 year	Two-thirds regained
RCT meta-analysis ¹¹	3,771	Median 21 weeks	Obesity: +5.63 kg
Newer-drug model ¹⁴	1,776	Up to 1 year	+0.8 kg/month
PCOS cohort ¹⁵	25	2 years	One-third regained

The 0.8 kg per month figure for newer drugs comes from ten semaglutide or tirzepatide arms; a return to baseline at about 1.5 years was a model projection because follow-up beyond one year was scarce.¹⁴ These are group averages, not a countdown clock.

03 Blood sugar, blood pressure, and lipids can drift back too

Semaglutide does more than alter the scale. During treatment it lowers glucose and improves several cardiovascular risk markers, and long trials show fewer major cardiovascular events and kidney outcomes in selected high-risk populations.^{16,17,18,19,20} Those outcome trials tested continued treatment. They do not prove that the protection persists after the medication is stopped.

Withdrawal studies mostly measure markers, not heart attacks or kidney failure. Across obesity trials, stopping a GLP-1 drug was followed by an average 0.25 percentage-point rise in HbA1c, a 0.45 mmol/L rise in fasting glucose, and a 4.15 mm Hg rise in systolic blood pressure; the estimates varied widely between studies.¹¹ In the STEP 1 extension, blood pressure returned to baseline by one year, while small residual improvements in HbA1c and some lipids remained.¹ Among people with type 2 diabetes, pooled HbA1c rose by 0.65 points after GLP-1 therapy stopped.¹¹

That distinction matters clinically. Someone taking Ozempic chiefly for diabetes can lose glucose control even if the scale changes little. In one specialized real-world program, HbA1c rose 0.4 points at six months and 0.6 at 12 months after deprescribing, despite no significant average weight regain.²¹ Conversely, STEP 10 showed how strongly active semaglutide can move prediabetes toward normal glucose;

One meta-analysis of eight randomized trials estimated 9.69 kg of regain after semaglutide or tirzepatide, with a wide 95% confidence interval of 5.78 to 13.60 kg.¹² Another synthesis of 18 randomized trials found 5.63 kg of average regain among people with overweight or obesity, with extreme between-study variation.¹¹ A newer nonlinear model estimated that 60% of lost weight returns by one year and that regain may slow below the original baseline, but its long-range curve is partly extrapolated.¹³

The studies below show why "you gain it all back" is too blunt.

once the drug is removed, part of that gain is liable to reverse.²²

Hard outcomes after discontinuation remain uncertain. In an older Veterans Affairs population with chronic kidney disease, gaps of at least 180 days in GLP-1 therapy were associated with higher rates of death, heart attack, heart-failure admission, and stroke.²³ But that observational association is vulnerable to confounding: people who stop treatment may be sicker, poorer, recently hospitalized, or less able to obtain care. Reviews therefore warn about lost protection without claiming that cessation itself has been proved to trigger an acute cardiovascular rebound.^{3,24}

04 The stomach often gets quieter, not "withdrawal sick"

Nausea, vomiting, diarrhea, constipation, and early fullness are drug effects, especially during dose escalation. Across randomized evidence, gastrointestinal events are much more common with semaglutide than placebo and are the leading reason people stop.^{25,26,27,28} In the 17,604-person SELECT trial, gastrointestinal events caused permanent discontinuation in 10.0% on semaglutide versus 2.0% on placebo, mostly during escalation.²⁹

Stopping usually removes that pressure rather than creating a new syndrome. In STEP 4, gastrointestinal events were reported by 49.1% of people who continued semaglutide and 26.1% of those switched to placebo.¹⁰ The trials did not identify a characteristic cluster of tremor, sweating, agitation, or other dependence-type symptoms. Appetite returning, however, can feel dramatic precisely because eating had required so little negotiation on treatment.

Persistent severe abdominal pain, repeated vomiting, dehydration, or symptoms that worsen after stopping are not something the withdrawal evidence explains away. They

warrant medical assessment rather than being assumed to be normal rebound.

05 The turn: regain is common, not destiny

Clinical trials deliberately isolate the drug effect. Real life is messier. In 7,881 patients, one-year weight loss averaged 11.9% among continuers, 6.8% among late discontinuers, and 3.6% among early discontinuers; prediabetes also improved less when treatment ended early.³⁰ Yet a later cohort of 7,938 discontinuers found only a 0.5% average weight increase in the following year among those treated for obesity - while 19.6% restarted the original drug and 35.2% received another obesity treatment.³¹ The small average therefore does not describe untreated cessation.

Selected maintenance studies offer clues, not recipes. Twenty-five women with polycystic ovary syndrome who continued metformin and lifestyle support regained about one-third of their semaglutide loss over two years.¹⁵ In the intensive remote-care cohort noted above, weight remained stable after deprescribing, but patients followed a carbohydrate-restricted program and usually continued metformin.²¹ These uncontrolled or observational results cannot establish that metformin, carbohydrate restriction, or telemedicine will prevent regain for the average person.

Large electronic-record cohorts confirm enormous individual variation and generally more modest weight change than efficacy trials.³² The honest answer is probabilistic: regain is the most common trajectory, full regain is not inevitable, and we still cannot reliably predict who will maintain most of the loss.

06 Stopping is often an access story - and sometimes temporary

People stop for many reasons besides a planned therapeutic decision. In one cohort, adverse reactions accounted for 26.8% of documented discontinuations and cost for 14.4%.³³ Older US claims data found 47.7% had discontinued injectable GLP-1 therapy by one year and 70.1% by two.³⁴ A UK diabetes cohort found similarly poor persistence, with 45.2% discontinued at one year and 64.7% at two.³⁵

Restarting is common. Among 125,474 US adults, 53.6% discontinued within one year; among eligible discontinuers, 47.3% with diabetes and 36.3% without it restarted within the next year. Each 1% of weight gained after stopping was associated with a 2.3% to 2.8% higher rate of restarting.³⁶ This cycle can reflect biology, side effects, shortages, insurance, and income all at once.

07 A safe stop is a handoff, not a cliff

No peer-reviewed randomized trial has established that tapering semaglutide prevents weight or glucose rebound. Recent reviews propose gradual reduction and structured transition plans, but they also acknowledge that validated tapering protocols are absent.^{3,24} Because semaglutide already clears slowly, tapering and stopping abruptly are

not pharmacologically as different as they would be for a short-acting drug - although a taper may still help some people test appetite, glucose, and weight responses in smaller steps.

The practical question is therefore not simply, "Can I stop?" It is, "What replaces each job the drug was doing?" For weight, that may mean a maintenance plan with food structure, resistance and aerobic activity, sleep, behavioral support, or another therapy. For diabetes, it means arranging glucose monitoring and an alternative medication plan before the last dose, because HbA1c can rise even without visible regain. Weight, waist, blood pressure, and glucose are more useful early-warning instruments than willpower.

The STEP 1 volunteers finished a year off semaglutide lighter than they began, but moving back toward their starting point. That is the clearest image to keep: stopping does not make the body malfunction. It removes a treatment that was actively changing appetite and metabolism. What happens next depends on the old biology, the size of the loss, and what - if anything - takes the drug's place.

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